

MULTAN ELECTRIC POWER COMPANY

Information Technology Directorate • Programming Internship Project

SOFTWARE REQUIREMENTS SPECIFICATION

MEPCO Integrated Help Desk & Ticket Management System

Information	Details
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Internship Period	13 July 2026 – 04 September 2026
Date	14 July 2026

Prepared for the MEPCO Information Technology Directorate internship project.

Project Overview

The MEPCO Integrated Help Desk & Ticket Management System is a centralized web application designed to serve two related service domains: external electricity consumers and internal MEPCO employees. Consumers submit line, non-line, billing, metering, safety, theft and service-request tickets. Employees submit departmental support tickets covering administrative, operational, technical, governance, finance, supply-chain, training and other internal needs. Both domains share a controlled ticket engine while retaining distinct identities, categories, fields, routing rules and visibility boundaries.

The system provides five roles—Consumer, Employee, Technician, Supervisor and Administrator. It supports secure registration and login, role-based access control, configurable ticket categorization, assignment and reassignment, progress tracking, comments and attachments, status history, notifications, reporting, audit trails, searchable paginated lists, and controlled administration. The project also includes a RESTful API, JWT access and refresh tokens, centralized validation/error handling, structured logging, Dockerized development, automated backend/API tests, Swagger/OpenAPI documentation, realistic seed data, and GitHub feature-branch and pull-request practices.

Simple project idea

The application works as two help desks inside one website: one for electricity consumers and one for MEPCO employees. Both use the same ticket system, while their accounts, complaint categories and access remain separate.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, data, interface, quality, security, operational and verification requirements for the MEPCO Integrated Help Desk & Ticket Management System. It is the agreed reference for stakeholder review, interface and database design, implementation planning, testing, demonstration and acceptance.

1.2 Product Scope

The product is a responsive browser-based help-desk platform for MEPCO. It enables external consumers and internal employees to register or authenticate, submit appropriately categorized tickets, provide supporting information, follow progress and respond to support staff. Departmental technicians process assigned work. Supervisors control routing, priority, reassignment, resolution review and operational reporting. Administrators manage users, master data, announcements, security settings and audit access.

- Consumer Services domain: public-facing electricity service complaints, safety reports, theft reports and service requests.
- Internal Employee Services domain: departmental support, operations, administration, finance, governance, IT and organizational requests.
- Shared platform services: identity, access control, ticket history, comments, attachments, notifications, search, reports, logging and audit.

1.3 Business Objectives

ID	Objective	Success Indicator
BO-01	Provide one traceable intake channel	Every submitted issue receives a unique ticket ID and status history.
BO-02	Improve accountability	Every active ticket has an owner or documented unassigned state; assignment changes are recorded.
BO-03	Increase transparency	Requesters can view permitted status, updates and resolution details without contacting staff separately.
BO-04	Support management oversight	Supervisors can view workload, aging, completion and technician performance summaries.
BO-05	Protect sensitive operations	RBAC, validation, audit logging and domain visibility controls apply to protected functions and data.
BO-06	Demonstrate sound engineering	The project includes API documentation, tests, Docker, seed data and GitHub review workflow.

1.4 Intended Audience

- MEPCO internship mentor, IT Directorate and reviewing management personnel.
- Consumer-service, operations and departmental representatives validating categories and workflows.
- Developer(s) implementing the React interface, Express API and MySQL database.
- Testers preparing unit, integration, API and acceptance tests.
- Future maintainers and administrators responsible for configuration and operation.

1.5 Document Conventions

Required behaviour uses the word “shall.” Recommendations or proposed defaults use “should” or are clearly labelled as assumptions. Each requirement has a short identifier so it can be discussed and tested easily. Consumer and Employee are collectively called requesters; Technician, Supervisor and Administrator are called support staff.

1.6 Definitions and Acronyms

Term	Definition
API	Application Programming Interface.
JWT	JSON Web Token used for access and refresh authentication.
RBAC	Role-Based Access Control.
SRS	Software Requirements Specification.
Ticket	A uniquely identified support complaint, incident, request, lead or other work item.
Consumer	An external MEPCO electricity customer represented by a billing Reference Number.
Employee	An internal MEPCO staff member represented by an Employee ID.
Technician	Departmental support personnel assigned to investigate and work on tickets.
Supervisor	Support manager who assigns/reassigns work, oversees progress and approves closure.
Audit log	Tamper-resistant record of security-relevant or business-critical actions.
Soft delete	Logical removal from normal views while preserving recoverable history and audit evidence.
SLA target	Configurable service target used for operational reporting; not a statutory commitment in this project baseline.

1.7 References

- Mentor-provided twelve-feature project brief (requirements incorporated in this SRS).
- MEPCO organizational departments and operational circle/city lists supplied for the project.
- Consumer and employee complaint catalogs supplied and expanded as configurable master data.

2. Overall Description

2.1 Product Perspective

The product is a new centralized system rather than an extension of a confirmed production platform. For the internship project it runs locally in a Dockerized environment and is designed so that a later production deployment can use the same application structure. It does not require paid services for development or demonstration.

Who Uses the System?

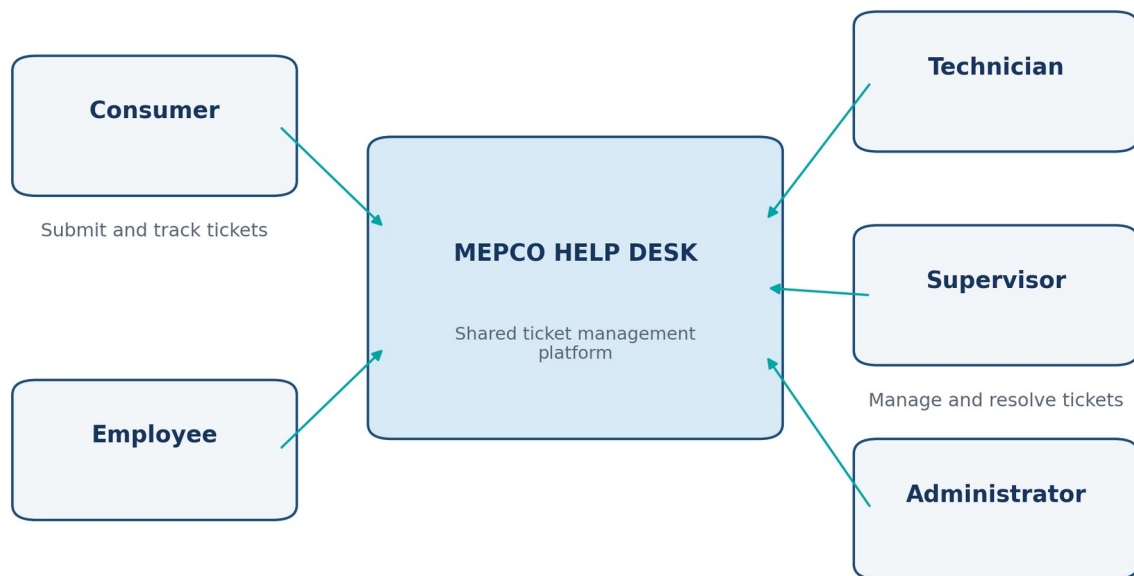


Figure 1 — System Context

2.2 Stakeholders and User Classes

Class	Primary Need	Key Characteristics / Access Boundary
Consumer	Submit and track electricity-service issues and requests.	Uses Reference Number; sees own profile and own tickets only.
Employee	Submit and track internal departmental issues.	Uses Employee ID; sees own profile and own tickets only; associated with department/location.
Technician	Work assigned tickets and document diagnosis/resolution.	Usually scoped to assigned tickets and supported departments/categories.
Supervisor	Control queues, assignments, priorities and operational reporting.	Views tickets within authorized service domains/departments; approves closure.
Administrator	Operate identity, configuration, audit and system controls.	Broad administrative access; all high-impact actions require audit evidence.
Project Mentor / Reviewer	Validate requirements and assess implementation.	Reviews documents, demonstration evidence, tests and repository practices.

2.3 Operating Environment

- Client: modern desktop and mobile browsers with JavaScript enabled; responsive layouts from small mobile screens to desktop monitors.
- Frontend: React with TypeScript and Vite.
- Backend: Node.js with Express and TypeScript, exposing JSON REST endpoints.
- Database: MySQL, selected to match existing developer familiarity and keep the project manageable.
- Development/verification: Windows or Linux host with Git, Node.js, Docker Desktop/Engine and Docker Compose.
- Documentation: Swagger/OpenAPI served by the backend; project documentation stored in the repository.

2.4 Constraints

ID	Constraint
CON-01	Development and demonstration shall use free/open tooling or free local capabilities; no paid cloud dependency is required.
CON-02	The project shall remain manageable for an internship and avoid unnecessary architectural complexity.
CON-03	MySQL shall be used; Prisma and Zod are excluded from the selected simplified implementation approach.
CON-04	The twelve mentor-mandated engineering features are mandatory and traceable.
CON-05	Consumer and employee records/tickets shall remain logically separated through domain/type fields and authorization checks.
CON-06	Production integration with authoritative MEPCO billing or employee directories is not assumed for the local internship demonstration.

2.5 Assumptions and Dependencies

- Reference Numbers and Employee IDs are unique within their respective identity domains.
- For the local demonstration, seed records or administrator-approved registration data represent authoritative identities.
- A supervisor or administrator configures which technicians support each department/category.
- Email/SMS gateways are not required for acceptance; in-application notifications are mandatory. External delivery may be added later.
- File storage may use a controlled local volume for the internship project; production storage can replace it later.
- Final complaint taxonomies, SLA targets and retention periods require MEPCO stakeholder validation and remain administrator-configurable.

2.6 Not Included in This Project

- Direct changes to billing, metering, ERP/SAP, HR, GIS, SCADA or grid-control systems.
- Automatic verification against live MEPCO consumer or employee databases.
- Native Android/iOS applications; the responsive web application is the delivery channel.
- Automated dispatch of field crews, route optimization, inventory reservation or financial settlement.
- Public social-media intake, chatbot/AI classification, predictive analytics or multilingual voice support.
- A legally binding SLA, regulatory complaint adjudication process, or production disaster-recovery guarantee.

3. System Design Overview

3.1 Architectural Principles

- Separation of concerns: presentation, API, application services, data access and persistence.
- Server-side authorization: the frontend may hide controls, but the API independently enforces all permissions.
- Configurable master data: departments, categories, priorities, statuses and announcements are managed rather than hard-coded where practical.
- Traceability by design: ticket history, assignments and audit records preserve who changed what and when.
- Shared core, separate domains: consumer and employee workflows reuse platform services while retaining distinct identity and classification rules.

Simple System Architecture

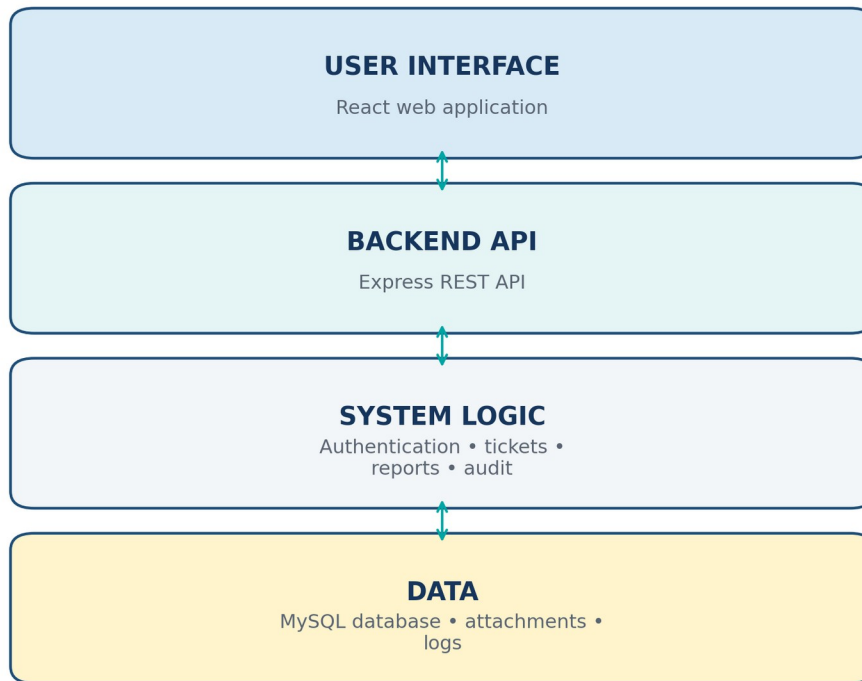


Figure 2 — Logical Architecture

3.2 Proposed Components

Component	Responsibility
React Web Client	Responsive role-specific routes, forms, dashboards, validation feedback and accessible interaction.
Express REST API	Authentication, authorization, validation, business rules, response envelopes and OpenAPI exposure.
Identity & RBAC Service	Registration/login, password handling, access/refresh tokens, session revocation and role/permission checks.
Ticket Service	Ticket creation, classification, assignment, transitions, history, comments, attachments and requester confirmation.
Notification Service	In-application notification creation, read/unread state and event-trigger mapping.
Reporting Service	Filtered operational summaries, queue aging, completion trends and export-ready data.
Audit Service	Append-only recording of security and administrative events, with authorized search.
MySQL Database	Transactional persistence for identities, configuration, tickets, history, assignments, notifications and audit metadata.
Controlled File Storage	Attachment bytes outside the database with validated metadata and access checks.
Docker Compose	Repeatable local startup for frontend, backend and database services.

3.3 Deployment View

The local Docker Compose setup contains frontend, API and MySQL services plus named volumes for database and attachment persistence. The browser communicates with the API over HTTP during local development. A future production environment should add HTTPS, protected secrets, scheduled backups and monitoring. Secrets shall never be committed to Git.

4. Functional Requirements

Each requirement has a short ID so it can be discussed, developed and tested easily. The tables focus only on what the system needs to do.

4.1 Identity, Registration and Authentication

Requirement ID	What the system must do
FR-AUTH-001	The system shall allow a Consumer to register using a unique MEPCO Reference Number, name, contact details, address, circle/city and password.
FR-AUTH-002	The system shall allow an Employee to register using a unique Employee ID, name, department, designation/contact details, work location and password.
FR-AUTH-003	Technician, Supervisor and Administrator accounts shall be created or activated by an authorized Administrator and shall authenticate using username and password.
FR-AUTH-004	The system shall prevent duplicate Reference Numbers, Employee IDs and staff usernames within their respective identity namespaces.
FR-AUTH-005	The system shall validate required registration fields and return field-level errors without exposing internal details.
FR-AUTH-006	Successful login shall issue a short-lived JWT access token and a longer-lived refresh token using secure rotation/revocation rules.
FR-AUTH-007	The system shall support access-token renewal using a valid, unexpired and non-revoked refresh token.
FR-AUTH-008	Logout shall revoke the current refresh session and prevent subsequent reuse of its token.
FR-AUTH-009	Repeated failed authentication attempts shall be rate-limited and recorded as security events.
FR-AUTH-010	An authorized Administrator shall be able to activate, suspend and reactivate accounts with a reason.
FR-AUTH-011	The system shall support an administrator-controlled password reset workflow; users shall never be shown an existing password.
FR-AUTH-012	The system shall enforce role-based route and endpoint authorization for every protected operation.

4.2 User Profiles and Master Identity Data

Requirement ID	What the system must do
FR-PROF-001	Authenticated users shall be able to view their own profile.
FR-PROF-002	Users shall be able to edit permitted contact and address fields while immutable identity keys remain protected.
FR-PROF-003	The system shall associate each Employee and staff account with an active department and location where applicable.

Requirement ID	What the system must do
FR-PROF-004	The system shall associate each Consumer with a Reference Number, circle and city/location.
FR-PROF-005	Authorized Administrators shall be able to search, filter and paginate user accounts by role, status, department and location.
FR-PROF-006	Changes to role, status, identity keys or departmental scope shall be audit-logged.

4.3 Ticket Creation and Classification

Requirement ID	What the system must do
FR-TKT-001	A Consumer or Employee shall be able to create a ticket within the correct service domain.
FR-TKT-002	Every submitted ticket shall receive a unique human-readable ticket number and creation timestamp.
FR-TKT-003	The creation form shall present dependent Category and Complaint Type choices appropriate to the requester domain.
FR-TKT-004	Employee ticket classification shall additionally support department-based routing.
FR-TKT-005	Consumer ticket classification shall capture circle, city and service location details as required.
FR-TKT-006	Every configurable dropdown shall include an Other choice that reveals a bounded free-text description where meaningful.
FR-TKT-007	A ticket shall contain a concise subject, detailed description, category/type, location/context, requester and initial status.
FR-TKT-008	A requester shall be able to add allowed attachments such as screenshots, bills or supporting documents at creation.
FR-TKT-009	The system shall prevent a requester from viewing or modifying another requester's ticket.
FR-TKT-010	A requester may edit a New/unassigned ticket's permitted fields; later edits shall occur through comments or authorized staff actions.
FR-TKT-011	A requester may withdraw a New ticket with a reason; the system shall preserve its history as Cancelled rather than physically deleting it.
FR-TKT-012	The API shall reject invalid domain/category/type combinations even if a client bypasses the interface.

4.4 Assignment, Queue Management and Priority

Requirement ID	What the system must do
FR-ASG-001	New tickets shall enter an authorized supervisor queue using service domain, department/category and location metadata.
FR-ASG-002	A Supervisor shall be able to assign a ticket to an eligible Technician.
FR-ASG-003	A Supervisor shall be able to reassign a ticket, recording the previous assignee, new assignee, actor, timestamp and reason.
FR-ASG-004	A Technician shall see tickets currently assigned to that technician within authorized scope.
FR-ASG-005	A Supervisor shall be able to set or change priority with a reason.

Requirement ID	What the system must do
FR-ASG-006	The system shall support Low, Medium, High and Critical priorities as configurable master data.
FR-ASG-007	Queue views shall support pagination, sorting and filters for status, priority, category, department, circle/city, technician and date range.
FR-ASG-008	Supervisors shall be able to identify unassigned, aging and overdue-against-target tickets.
FR-ASG-009	Technicians shall not self-assign outside their authorized scope unless an Administrator explicitly enables that policy.

4.5 Ticket Processing, Communication and Resolution

Requirement ID	What the system must do
FR-WRK-001	An assigned Technician shall be able to transition a ticket from Assigned to In Progress.
FR-WRK-002	A Technician shall be able to add public comments visible to the requester.
FR-WRK-003	Authorized support staff shall be able to add internal notes not visible to Consumer or Employee requesters.
FR-WRK-004	A Technician shall be able to request information from the requester and set the ticket to Pending User.
FR-WRK-005	A requester response to Pending User shall be recorded and return the ticket to an actionable support state.
FR-WRK-006	A Technician shall be able to attach authorized solution or evidence files.
FR-WRK-007	A Technician shall propose resolution by entering a mandatory resolution summary and setting the ticket to Resolved.
FR-WRK-008	A Supervisor shall be able to approve a valid resolution and close the ticket, or return it for additional work with a reason.
FR-WRK-009	A requester shall be able to confirm satisfaction/closure or request reopening with a reason while the configured reopening window remains open.
FR-WRK-010	A Supervisor shall be able to reopen a Closed ticket when authorized, preserving the complete history.
FR-WRK-011	Every status change shall record old status, new status, actor, timestamp and optional/required reason.
FR-WRK-012	Ticket conversations shall display in chronological order with author, visibility and timestamp.

4.6 Notifications and Announcements

Requirement ID	What the system must do
FR-NOT-001	The system shall create in-application notifications for ticket creation, assignment, reassignment, comments, information requests, resolution, closure and reopening as applicable.
FR-NOT-002	Users shall be able to view unread/read notifications and mark individual or all notifications as read.
FR-NOT-003	A notification shall link to the authorized ticket or relevant page without bypassing access control.
FR-NOT-004	Administrators shall be able to publish, schedule/expire, activate and deactivate announcements for selected audiences.

Requirement ID	What the system must do
FR-NOT-005	External email/SMS delivery may be added through adapters without changing the core ticket model.

4.7 Search, Dashboards and Reports

Requirement ID	What the system must do
FR-REP-001	Each role shall have a dashboard showing only permitted and relevant counts, queues and recent activity.
FR-REP-002	Authorized ticket lists shall support keyword search across ticket number, subject and permitted text fields.
FR-REP-003	List endpoints shall support validated pagination, sorting and filtering with safe defaults and maximum page sizes.
FR-REP-004	Supervisors shall be able to view tickets created, assigned, resolved, closed, reopened and pending within a date range.
FR-REP-005	Supervisors shall be able to view technician workload and completion summaries within authorized scope.
FR-REP-006	Reports shall support breakdowns by domain, category/type, department, circle/city, priority, status and technician where applicable.
FR-REP-007	Administrators shall be able to export permitted report data to CSV without exporting password, token or secret values.
FR-REP-008	Daily operational summaries shall show completed versus incomplete work and aging queues.

4.8 Administration and Configuration

Requirement ID	What the system must do
FR-ADM-001	Administrators shall manage users, roles and permitted account states.
FR-ADM-002	Administrators shall manage departments, circles, cities, categories, complaint types, priorities and status configuration.
FR-ADM-003	Master-data records referenced by tickets shall be deactivated rather than physically deleted.
FR-ADM-004	Administrators shall manage technician support scopes and supervisor authorization scopes.
FR-ADM-005	Administrators shall view searchable audit logs subject to least-privilege access.
FR-ADM-006	Administrators may correct or soft-delete an erroneous ticket only with a mandatory reason; the action shall be recoverable and audited.
FR-ADM-007	Administrators shall be able to execute or document approved seed-data loading for local environments.
FR-ADM-008	The system shall prevent an administrator from removing the last active Administrator account.
FR-ADM-009	High-impact changes shall require confirmation and return a clear success or error result.

4.9 Audit and History

Requirement ID	What the system must do
FR-AUD-001	The system shall maintain append-only audit records for authentication events, role/status changes, master-data changes, assignments, administrative ticket correction and other configured sensitive actions.
FR-AUD-002	Each audit record shall include actor, action, target entity/identifier, timestamp, result and relevant before/after metadata without storing plaintext secrets.
FR-AUD-003	Authorized users shall be able to filter audit logs by actor, action, entity, result and date range.
FR-AUD-004	Normal application users shall not be able to update or delete audit records through application endpoints.
FR-AUD-005	Ticket history shall be visible to authorized ticket participants and shall exclude confidential audit-only metadata.

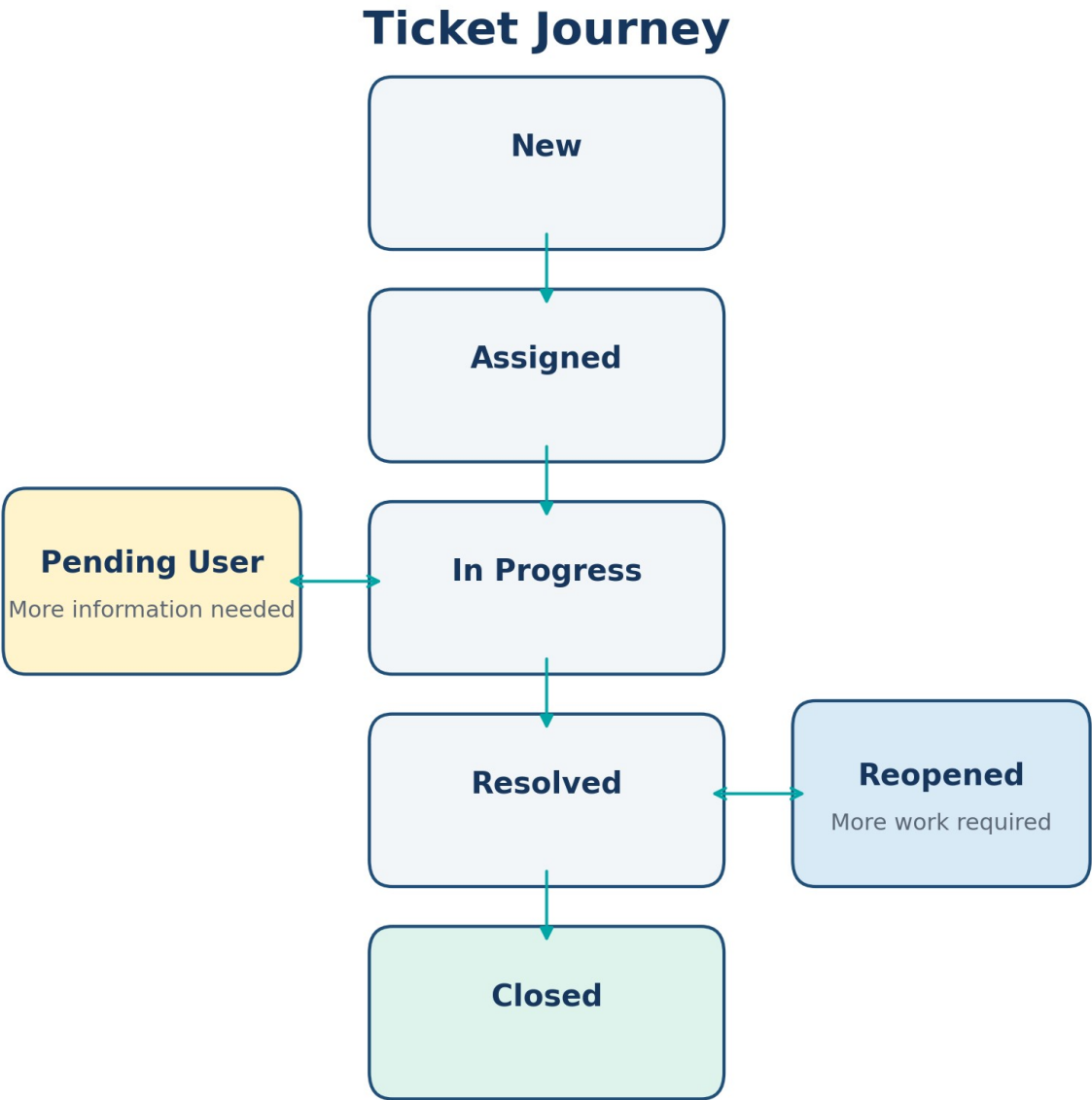
5. Roles, Permissions and Workflow Rules

5.1 What Each Role Can Do

Role	Main permissions
Consumer	Create and view personal consumer complaints, add comments/files, follow progress, and confirm or request reopening of a resolution.
Employee	Create and view personal internal tickets, add comments/files, follow progress, and confirm or request reopening of a resolution.
Technician	View assigned tickets, start work, add public updates and internal notes, request information, and submit a resolution.
Supervisor	View authorized queues, assign or reassign technicians, change priority, monitor progress, approve closure, reopen tickets, and view reports.
Administrator	Manage accounts and system lists, configure access, view audit logs, correct erroneous tickets with a reason, and view system-wide reports.

Users can only access records allowed by their role. The backend checks every protected action even when a button is hidden in the interface.

5.2 Ticket Lifecycle



Every change is saved in the ticket history.

Figure 3 — Ticket Status Lifecycle

Status	Meaning	Typical Allowed Next States
New	Submitted and awaiting triage/assignment.	Assigned; Cancelled
Assigned	Ownership given to an eligible technician.	In Progress; Assigned (reassigned); Cancelled
In Progress	Technician is actively investigating or working.	Pending User; Resolved; Assigned (reassigned)
Pending User	Requester information/confirmation is required.	In Progress; Assigned; Cancelled by authorized rule

Status	Meaning	Typical Allowed Next States
Resolved	Technician supplied a resolution awaiting confirmation/approval.	Closed; In Progress; Reopened
Closed	Work is accepted/approved and complete.	Reopened within authorized policy
Reopened	A valid recurrence or rejected resolution requires more work.	Assigned; In Progress; Resolved
Cancelled	Withdrawn, duplicate, invalid or administratively cancelled with reason.	Reopened by Supervisor/Admin only

5.3 Priority Model

Priority	When it is used	Examples
Critical	Immediate danger, very large outage or serious security risk.	Live fallen wire, electric fire, major outage.
High	Major impact and no reasonable workaround.	Transformer failure or important system outage.
Medium	Important issue affecting a limited number of people.	Billing, meter or departmental workflow problem.
Low	Routine request or low-impact issue.	Information request or office-supply problem.

Safety routing

Critical safety complaints must be visually distinguished and routed for immediate human attention. This software records and communicates the issue; it does not replace emergency or field-safety procedures.

6. Main Use Cases

6.1 UC-01 — Consumer Registration

Element	Specification
Primary actor(s)	Consumer
Preconditions	No active account exists for the Reference Number.
Main success flow	Enter Reference Number and profile data → validate uniqueness/fields → create account → authenticate or proceed to login.
Postconditions	Consumer account exists; registration event is logged.
Exceptions / alternate paths	Duplicate identity; invalid required data; suspended reference record.

6.2 UC-02 — Employee Registration

Element	Specification
Primary actor(s)	Employee

Element	Specification
Preconditions	No active account exists for the Employee ID.
Main success flow	Enter Employee ID, department/location, contact and password → validate → create account.
Postconditions	Employee account exists and is associated with master data.
Exceptions / alternate paths	Duplicate ID; inactive department; invalid data.

6.3 UC-03 — Submit Ticket

Element	Specification
Primary actor(s)	Consumer / Employee
Preconditions	Requester is authenticated and active.
Main success flow	Choose domain-appropriate category/type → enter subject/details/location → add optional files → submit.
Postconditions	Unique New ticket and history record exist; requester/supervisor notifications are created.
Exceptions / alternate paths	Invalid category pairing; blocked file; missing field; duplicate submission retry.

6.4 UC-04 — Assign or Reassign Ticket

Element	Specification
Primary actor(s)	Supervisor
Preconditions	Ticket is visible within supervisor scope.
Main success flow	Open queue → inspect classification/priority → select eligible technician → provide reason when reassigning → confirm.
Postconditions	Current assignment changes; complete assignment history and notification exist.
Exceptions / alternate paths	Technician inactive/ineligible; stale concurrent update; unauthorized scope.

6.5 UC-05 — Process Ticket

Element	Specification
Primary actor(s)	Technician
Preconditions	Ticket is assigned and technician is authorized.
Main success flow	Start work → add public update/internal note/files → optionally request information → continue investigation.
Postconditions	Ticket status, conversation and history reflect work performed.
Exceptions / alternate paths	Unauthorized ticket; invalid transition; blocked file.

6.6 UC-06 — Resolve and Close

Element	Specification
Primary actor(s)	Technician + Supervisor / Requester
Preconditions	Investigation is complete.
Main success flow	Technician enters resolution → sets Resolved → supervisor/requester reviews → approve/confirm Close or return/reopen.
Postconditions	Ticket is Closed with resolution, closure actor and timestamps, or returned to active work.
Exceptions / alternate paths	Missing resolution; unauthorized closer; reopening window/policy failure.

6.7 UC-07 — Generate Daily Report

Element	Specification
Primary actor(s)	Supervisor
Preconditions	Supervisor is authenticated with reporting scope.
Main success flow	Choose date/scope/filters → system aggregates created, completed, pending, aging and technician workload → display/export if allowed.
Postconditions	Accurate, scoped report is displayed without exposing unauthorized data.
Exceptions / alternate paths	Invalid range; no data; export failure.

6.8 UC-08 — Manage Master Data

Element	Specification
Primary actor(s)	Administrator
Preconditions	Administrator is authenticated.
Main success flow	Choose master-data type → add/edit/deactivate item → confirm → system validates references and records audit event.
Postconditions	Configuration changes safely without breaking historical tickets.
Exceptions / alternate paths	Duplicate value; referenced record deletion attempt; last required item protection.

7. Data Requirements

7.1 Conceptual Data Model

Main Information Stored

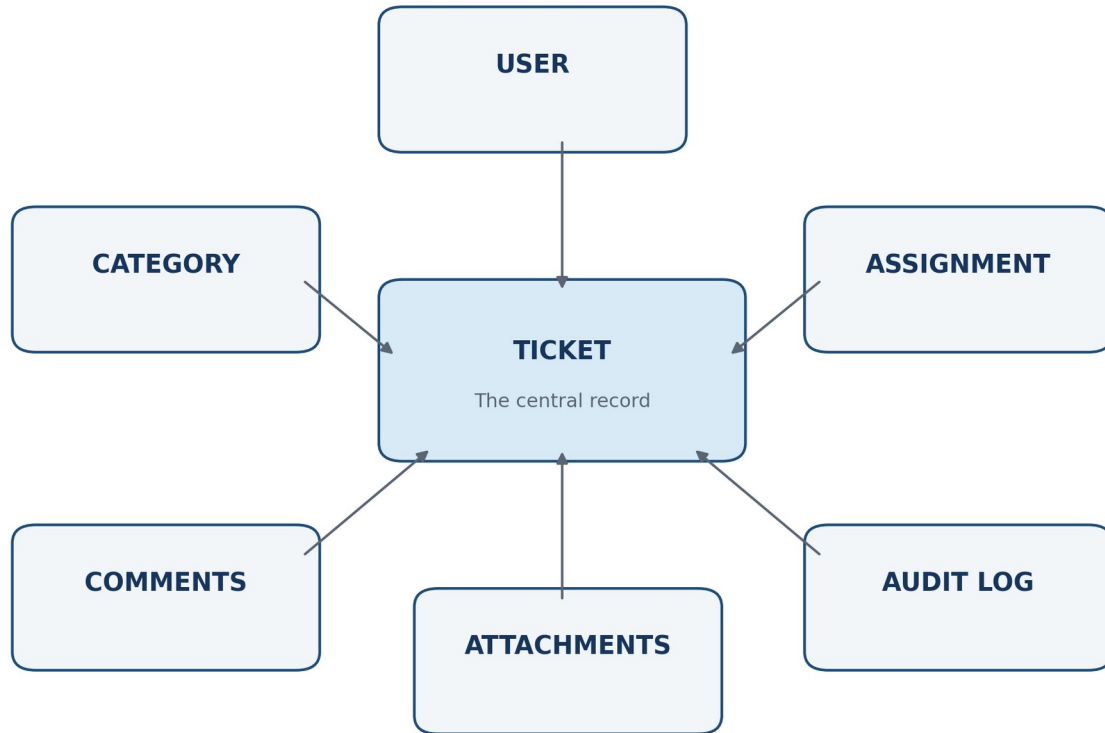


Figure 4 — Conceptual Data Model

7.2 Core Data Dictionary

Entity	Purpose	Representative Fields / Rules
User	Identity and account state for all roles.	user_id; role_id; display_name; email/phone; password_hash; status; timestamps. Identity-type-specific key is unique and nullable only when not applicable.
Consumer Profile	External consumer attributes.	user_id; reference_number; address; circle_id; city_id; optional service-location details.
Employee Profile	Internal staff attributes.	user_id; employee_id; department_id; designation; work location.
Role / Permission	Authorization definitions.	role_id; role_name; permission keys; active status.
Department	MEPCO organizational routing masters.	department_id; name; description; active; sort_order.
Circle / City	Consumer geographical routing masters.	circle_id; city_id; names; parent relationship; active.
Category / Complaint Type	Domain and departmental classification.	category_id; parent_id/type_id; domain; department_id; active; display order.

Entity	Purpose	Representative Fields / Rules
Ticket	Current state and core requester issue.	ticket_id; ticket_number; requester_id; domain; subject; description; category/type; priority; status; location; current_assignee; version; timestamps.
Ticket History	Immutable business-state timeline.	history_id; ticket_id; event_type; old/new values; actor_id; reason; timestamp.
Assignment	Complete ownership changes.	assignment_id; ticket_id; technician_id; assigned_by; reason; start/end timestamps.
Comment	Public or internal communication.	comment_id; ticket_id; author_id; visibility; body; timestamp; edit policy.
Attachment	Validated file metadata and storage reference.	attachment_id; ticket/comment owner; original/stored name; MIME; size; checksum/path; uploader; timestamp.
Notification	Role/user event alert.	notification_id; recipient_id; type; message; target; read_at; created_at.
Announcement	Audience-targeted notices.	announcement_id; title; body; audience; active/start/end; author.
Refresh Session	Revocable login session.	session_id; user_id; token_hash/id; expiry; revoked_at; device/IP metadata as approved.
Audit Log	Append-only security/administrative evidence.	audit_id; actor; action; entity; entity_id; result; before/after metadata; timestamp.

7.3 Ticket Fields

Information	Consumer Ticket	Employee Ticket
Requester	Consumer account and Reference Number	Employee account and Employee ID
Classification	Consumer complaint category and type	Department and internal complaint type
Location	Circle, city and service address	Department and work location
Issue details	Subject and full description	Subject and full description
Ticket control	Priority, status and assigned technician	Priority, status and assigned technician
Updates	Comments and attachments	Comments, attachments and internal staff notes
Completion	Resolution summary and confirmation	Resolution summary and confirmation

7.4 Integrity, Retention and Recovery

Requirement ID	What the system must do
DR-001	The database shall enforce unique identity keys and foreign-key relationships for core entities.
DR-002	Ticket creation and each state-changing operation shall use transactions where partial updates would create inconsistency.
DR-003	Historical tickets shall retain readable snapshot values when related master data is later renamed or deactivated.
DR-004	Passwords, refresh-token secrets and application secrets shall never be stored in plaintext.

Requirement ID	What the system must do
DR-005	Local development data shall persist through named Docker volumes unless intentionally reset.
DR-006	A documented database backup and restore procedure shall be provided and tested for the demonstration environment.
DR-007	Production retention and deletion periods shall be configurable or documented after MEPCO approval; this SRS does not invent a regulatory period.

8. External Interface Requirements

8.1 User Interface

Requirement ID	What the system must do
UI-001	The interface shall be responsive and usable on mobile, tablet and desktop viewport sizes.
UI-002	Role-specific navigation shall expose only relevant functions while preserving server-side authorization.
UI-003	Forms shall show labels, required indicators, inline errors and a clear submission result.
UI-004	Dependent dropdowns shall update Category/Complaint Type and Circle/City choices without losing unrelated user input.
UI-005	Status and priority shall use consistent text labels and color cues; color shall not be the only means of identification.
UI-006	Potentially destructive or high-impact actions shall require explicit confirmation and a reason where required.
UI-007	Loading, empty, success and failure states shall be clearly represented on lists and dashboards.
UI-008	Keyboard navigation, visible focus, semantic headings, labelled controls and reasonable contrast shall be supported.

8.2 Software Interfaces

Interface	Requirement
React ↔ Express	JSON over versioned REST endpoints; Bearer access token for protected requests; standardized response/error envelope.
Express ↔ MySQL	Parameterized queries through a maintained MySQL driver/data-access layer; pooled connections and transactional operations.
API Documentation	Swagger/OpenAPI UI generated from or synchronized with endpoint schemas and examples.
File Storage	Application-controlled interface storing validated files outside public web roots and serving them only after authorization.
GitHub	Repository, issues/branches/commits, pull requests and review history; no runtime dependency.
Future gateways	Email/SMS can be added later but are not required for the local project.

8.3 Communications Interfaces

- Local development may use HTTP on loopback or trusted development networks.
- Production deployment shall use HTTPS and secure cookie/header settings where applicable.
- API requests and responses shall use UTF-8 JSON; file uploads shall use controlled multipart requests.
- CORS shall permit only configured frontend origins rather than unrestricted production access.

9. REST API Requirements

9.1 API Conventions

Requirement ID	What the system must do
API-001	The backend shall expose resource-oriented REST endpoints under a version prefix such as /api/v1.
API-002	Successful responses shall use a consistent envelope containing success, data and optional message/meta fields.
API-003	Error responses shall use a consistent envelope containing success=false, stable error code, safe message and optional field details/request ID.
API-004	Endpoints shall return appropriate HTTP status codes including 200/201/204, 400, 401, 403, 404, 409, 422, 429 and 500 as applicable.
API-005	All request path, query, body and file inputs shall be validated before business processing.
API-006	Pagination metadata shall include page, pageSize, totalItems and totalPages; page size shall have a configured upper bound.
API-007	Sorting shall use an allow-list of fields and directions; filtering shall use typed validated parameters.
API-008	Unhandled errors shall be processed by centralized error middleware and shall not leak stack traces in production mode.
API-009	OpenAPI documentation shall describe authentication, schemas, parameters, responses and representative examples for all acceptance endpoints.

9.2 Resource Summary

Resource	Representative Operations	Primary Access
/auth	register consumer/employee; login; refresh; logout; reset password	Public + authenticated session
/users	own profile; admin list/view/status/role changes	Self / Administrator
/tickets	create; own/authorized lists; detail; update permitted fields; transition	All roles within scope
/tickets/{id}/assignments	assign/reassign and read history	Supervisor / Administrator
/tickets/{id}/comments	public comments; internal notes	Authorized participants
/tickets/{id}/attachments	upload/list/download/delete permitted file	Authorized participants
/notifications	list, unread count, mark read	Authenticated recipient
/reports	summary, aging, technician workload, category/location trends, export	Supervisor / Administrator

Resource	Representative Operations	Primary Access
/admin/master-data	departments, circles, cities, categories, complaint types, priorities/statuses	Administrator
/audit-logs	search/filter/detail	Authorized Administrator
/announcements	active audience view; admin CRUD/activation	Authenticated / Administrator
/health	service readiness/liveness without secrets	Operations / local verification

9.3 Example Response Contracts

Success: { "success": true, "message": "Ticket created", "data": { "ticketNumber": "MEPCO-2026-000123" }, "meta": null }

Error: { "success": false, "error": { "code": "VALIDATION_ERROR", "message": "Check the highlighted fields", "details": [...] }, "requestId": "..." }

10. Non-Functional Requirements

10.1 Security and Privacy

Requirement ID	What the system must do
NFR-SEC-001	Passwords shall be processed using a current adaptive one-way password hashing algorithm with unique salts.
NFR-SEC-002	Access tokens shall be short-lived; refresh sessions shall support rotation, expiry and revocation.
NFR-SEC-003	Every protected API operation shall authenticate the user and authorize role, domain and record scope.
NFR-SEC-004	Database operations shall use parameterized queries and output shall be safely encoded to reduce injection and XSS risk.
NFR-SEC-005	Sensitive configuration shall be provided through environment variables or managed secrets and excluded from source control.
NFR-SEC-006	File uploads shall enforce allow-listed MIME/extensions, maximum size, generated storage names and authorization on download.
NFR-SEC-007	Authentication and mutation endpoints shall use appropriate rate limits and security headers.
NFR-SEC-008	Logs, reports and API responses shall not expose password hashes, raw refresh tokens, secrets or unnecessary personal data.
NFR-SEC-009	Production transport shall use HTTPS; local HTTP is permitted only for controlled development.
NFR-SEC-010	Security-relevant operations shall be audit logged with a request/correlation identifier where practical.

10.2 Performance and Capacity

Requirement ID	What the system must do
NFR-PERF-001	Under the local acceptance workload, 95% of normal non-report API requests should complete within 2 seconds excluding network and file-transfer time.
NFR-PERF-002	Filtered ticket lists of up to the seeded acceptance dataset shall return the first page within 3 seconds under normal local conditions.

Requirement ID	What the system must do
NFR-PERF-003	The system shall use pagination and shall not return unbounded user, ticket, notification or audit collections.
NFR-PERF-004	Frequently filtered database columns shall be indexed based on measured query plans.
NFR-PERF-005	The baseline shall support at least 50 concurrent simulated authenticated sessions for demonstration testing without data corruption.

10.3 Reliability and Availability

Requirement ID	What the system must do
NFR-REL-001	A failed multi-step mutation shall roll back rather than leave partial ticket, assignment or history data.
NFR-REL-002	Duplicate client retries shall not silently create duplicate tickets where an idempotency mechanism or duplicate-submission guard is applied.
NFR-REL-003	The application shall expose a non-sensitive health endpoint for local readiness verification.
NFR-REL-004	The project shall document clean startup, shutdown, backup and restore procedures.

10.4 Usability and Accessibility

Requirement ID	What the system must do
NFR-USA-001	A first-time requester shall be able to register/login and submit a valid ticket without training using clear labels and feedback.
NFR-USA-002	Common role tasks shall be reachable through consistent navigation and role dashboards.
NFR-USA-003	The interface shall support keyboard use, visible focus, semantic labels and non-color-only status communication.
NFR-USA-004	Dates, times and identifiers shall use consistent formats; stored timestamps shall have an unambiguous timezone strategy.

10.5 Maintainability, Portability and Compatibility

Requirement ID	What the system must do
NFR-MNT-001	Frontend, backend and data-access modules shall have clear responsibilities and consistent naming.
NFR-MNT-002	Environment-specific values shall be configurable without source-code edits.
NFR-MNT-003	Database schema creation/migrations and seed execution shall be repeatable and documented.
NFR-MNT-004	The application shall run through Docker Compose on supported developer hosts with documented prerequisites.
NFR-MNT-005	The responsive client shall support current major Chromium- and Firefox-based desktop browsers and common mobile browser layouts used for testing.
NFR-MNT-006	Public functions and API contracts shall be documented sufficiently for another developer to continue the project.

11. Logging, Error Handling and Audit Requirements

Area	Requirement
Application logging	Use structured logs with timestamp, level, service, request ID, route, result and safe diagnostic context.
Log levels	At minimum: error, warn, info and debug; debug disabled or restricted in production mode.
Centralized errors	One backend error pipeline maps validation, authentication, authorization, not-found, conflict and unexpected errors to consistent responses.
Audit separation	Business/security audit events are distinct from diagnostic logs and use more restrictive access/retention controls.
Privacy	Passwords, tokens, secrets, uploaded contents and unnecessary personal data are redacted or excluded.
Correlation	A request ID connects API errors, diagnostic logs and relevant audit entries without revealing secrets.
Failure behavior	Unexpected failures return a safe generic message while detailed diagnostics remain in protected logs.

12. Testing and Verification Requirements

Requirement ID	What the system must do
TST-001	Backend business rules and reusable service functions shall have automated unit tests.
TST-002	REST endpoints shall have automated integration/API tests using an isolated test database.
TST-003	Authentication tests shall cover login, refresh rotation, logout/revocation, suspended accounts and invalid credentials.
TST-004	RBAC tests shall include permitted and forbidden operations for all five roles and cross-record/cross-domain attempts.
TST-005	Ticket workflow tests shall cover valid and invalid transitions, assignment history, resolution and reopening.
TST-006	Validation tests shall cover required fields, boundaries, invalid enum/master data, unsafe input and file restrictions.
TST-007	List endpoint tests shall verify pagination, filtering, sorting, maximum page size and authorization scope.
TST-008	Audit tests shall confirm required sensitive actions are recorded and ordinary users cannot modify audit data.
TST-009	A clean-environment acceptance run shall start with Docker Compose, apply schema/seed data and execute critical user journeys.
TST-010	Swagger/OpenAPI endpoints and examples shall be reviewed against implemented behavior before handoff.

12.1 Minimum Acceptance Scenarios

Scenario	Expected Result
Consumer creates Critical live-wire complaint	Correct consumer catalog/location captured; unique ticket created; urgent visual/routing indication; history/notifications recorded.
Employee submits IT hardware issue	Employee/department identity captured; ticket routed to authorized queue; technician can be assigned.
Unauthorized requester opens another ticket	API returns 403/404 according to disclosure policy; no protected data is leaked; attempt may be logged.
Supervisor reassigns active ticket	Old/new assignee, actor, reason and timestamps appear in assignment history; recipients notified.
Technician resolves without summary	Validation rejects transition and leaves the prior state unchanged.
Admin deactivates used category	Historical tickets remain readable; category disappears from new-ticket active choices.
Refresh token is reused after rotation/logout	Request is rejected and relevant security event is recorded.
Docker environment is rebuilt	Documented commands produce an operational seeded system without manual database editing.

13. Development, GitHub and Docker Requirements

Area	Requirement
Repository	One GitHub repository with README, documentation, frontend/backend source, environment examples, schema/migrations, seeds, tests and Docker configuration.
Branches	Feature branches use a consistent convention (for example feature/ticket-create, fix/auth-refresh); protected main remains releasable.
Pull requests	Changes reach main through pull requests with summary, testing evidence and linked requirement/issue where practical.
Commits	Commits are focused, descriptive and exclude generated secrets, local databases, dependency folders and private uploads.
Review	At least self-review/checklist or mentor/peer review evidence is retained before merge.
Docker	Compose defines frontend, backend and MySQL, health/dependency behavior, ports, environment variables and named volumes.
Configuration	.env.example documents required keys with non-secret examples; real .env is ignored.
README	Documents prerequisites, setup, development, tests, seed/reset, Swagger URL, backup/restore and troubleshooting.

14. Seed Data Requirements

Seed scripts shall produce a realistic but fictional demonstration dataset. They must be repeatable, avoid real consumer/employee personal data, and make acceptance scenarios easy to demonstrate.

Seed Area	Minimum Content
Users	At least one active account for each role, plus inactive/suspended examples; documented non-production credentials.
Organization	All departments, circles and listed cities including Other options.
Catalogs	Complete consumer catalog and representative department-specific employee catalog from the appendices.
Tickets	Consumer and employee tickets across all statuses/priorities, including unassigned, aging, resolved and reopened examples.
Activity	Assignments, comments, internal notes, notifications and audit events sufficient to demonstrate timelines/reports.
Safety	No real billing references, employee IDs, phone numbers, addresses or secret values.

15. How the Required Features Are Covered

Required project feature	Covered in this SRS	How it will be shown
Mentor 1: GitHub repository with feature branches and PRs	Section 13	Repository/PR review
Mentor 2: RESTful API with consistent response formats	FR groups; API-001-009	Contract/API tests
Mentor 3: JWT authentication with refresh tokens	FR-AUTH-006-009; NFR-SEC	Authentication/security tests
Mentor 4: Role-based access control	FR-AUTH-012; Section 5	Five-role authorization suite
Mentor 5: Input validation and centralized error handling	FR-AUTH-005; API-003-008; Section 11	Validation/error tests
Mentor 6: Logging and audit trails	FR-AUD; Section 11	Log/audit inspection and tests
Mentor 7: Pagination, sorting and filtering	FR-ASG-007; FR-REP-003; API-006-007	List endpoint tests
Mentor 8: Responsive React UI	UI-001-008	Viewport/usability review
Mentor 9: Dockerized development environment	NFR-MNT-004; Section 13	Clean startup acceptance
Mentor 10: Unit tests for backend logic and API	TST-001-009	Automated test report
Mentor 11: Swagger/OpenAPI documentation	API-009; TST-010	Documentation review
Mentor 12: Realistic seed scripts	DR-005; Section 14	Seed/reset demonstration

15.1 Project Completion Checklist

- The main functional and quality requirements are implemented or clearly discussed with the mentor.
- Critical Consumer and Employee end-to-end journeys pass in a clean Docker environment.
- All five roles pass positive and negative authorization tests.
- Automated backend/API tests pass and key evidence is included in the repository or final report.
- Swagger/OpenAPI matches the implemented acceptance endpoints.
- Seed data demonstrates departments, locations, catalogs, workflows, reports, notifications and audit trails.
- No known Critical or High security/data-integrity defect remains open.

- The SRS, architecture/design notes, setup guide and presentation are updated to reflect the delivered system.

16. Project Risks and Decisions

ID	Risk / Open Decision	Impact	Mitigation / Required Action
R-01	Final confirmation of the combined consumer + employee scope	High	Obtain mentor approval; architecture already isolates domains so either can be reduced without redesign.
R-02	No live identity/billing directory integration for local project	Medium	Use fictional seed/approved accounts; document integration boundary for future work.
R-03	Complaint catalog may differ from current operational taxonomy	Medium	Keep master data configurable; validate appendices with departmental stakeholders.
R-04	Sensitive HR, corruption, audit and intelligence complaints	High	Add category-level visibility restrictions in future refinement; default to least privilege and audited access.
R-05	Attachment malware or confidential data	High	Restrict type/size, isolate storage, authorize downloads; add antivirus scanning in production.
R-06	Over-expansion during internship	High	Implement the main required features first; external gateways, AI and native apps remain out of scope.
R-07	Undefined SLA and retention values	Medium	Use configurable targets and obtain formal MEPCO values before production use.
R-08	Single-developer quality risk	Medium	Use feature branches, pull-request checklist, automated tests, linting and frequent mentor reviews.

Stakeholder validation gate

Before implementation is treated as production-ready, MEPCO must validate scope, complaint taxonomy, confidential-case access, identity verification, escalation targets, data retention and deployment/security policies.

17. Future Enhancements

- Integration with authoritative consumer billing/reference and employee directory systems.
- Email, SMS or WhatsApp notification gateways subject to policy and consent.
- Urdu/localization support and richer accessibility testing.
- Configurable SLA escalation engine, duty rosters and field-team dispatch integration.
- Knowledge base, suggested solutions and duplicate-ticket detection.
- Approved analytics dashboards and geographical visualization.
- Production-grade malware scanning, object storage, centralized observability and disaster recovery.

Appendix A — MEPCO Departments

The following initial master data is supplied for the project. Every relevant selection includes Other, and administrators may add/deactivate items without deleting historical references.

Department / Unit	Purpose / Routing Context
Chief Executive Office (CEO)	Oversees macro corporate strategy.

Department / Unit	Purpose / Routing Context
Company Secretary Office	Handles board governance and legal matters.
Internal Audit Section	Reviews compliance and asset security.
HR & Administration Directorate	Coordinates public relations, legal/labor affairs, transport, and testing.
Finance Directorate	Oversees general accounts, pricing, and revenue collection.
Commercial/Customer Services Directorate (CSD)	Directs billing, customer service, meters, safety metrics, and testing (M&T).
Operations (OP) Directorate	Manages power distribution networks, feeders, and field staff.
Information Technology (IT) Directorate	Develops online applications, MIS frameworks, and net-metering databases.
Strategic Planning & Engineering Directorate	Governs network expansions and upgrades.
Operations & Maintenance (O&M) T&G Directorate	Secures and regulates transmission infrastructure and grid operations.
Supply Chain Management Office	Procures materials, hardware, and structural warehouse units.
Regional Training Centre (RTC)	Directs engineering technical courses and employee development programs.
MEPCO Intelligence/Surveillance Unit	Investigates electricity theft, illegal bypasses, and line tampering.
Other	Captures an organizational unit not yet present in the configurable master list.

Appendix B — Operational Circles and Cities

Circle	Cities / Locations
Multan Circle	Multan; Shujabad; Jalalpur Pirwala; Other
Khanewal Circle	Khanewal; Kabirwala; Mian Channu; Jahanian; Other
Vehari Circle	Vehari; Burewala; Mailsi; Other
Sahiwal Circle	Sahiwal; Chichawatni; Pakpattan; Arifwala; Other
Dera Ghazi Khan (D.G. Khan) Circle	Dera Ghazi Khan; Taunsa Sharif; Kot Chutta; Rajanpur; Jampur; Rojhan; Other
Muzaffargarh Circle	Muzaffargarh; Kot Addu; Alipur; Khan Garh; Other
Layyah Circle	Layyah; Chowk Azam; Keoror Lal Esan; Other
Bahawalpur Circle	Bahawalpur; Ahmedpur East; Hasilpur; Lodhran; Kahrora; Duniyapur; Other
Bahawalnagar Circle	Bahawalnagar; Chishtian; Haroonabad; Fort Abbas; Other
Rahim Yar Khan Circle	Rahim Yar Khan; Sadiqabad; Liaquatpur; Khanpur; Other
Other	Location not currently represented in the configurable circle/city list.

Appendix C — Consumer Complaint Catalog

The catalog is configurable. Spelling is preserved from the supplied project list where practical. Category/type validation occurs on both client and server.

C.1 Line Complaints

No.	Complaint Type
1	Power Outage
2	Damaged Meter
3	Electric Fire
4	Fluctuation
5	Line Fault
6	Low/ High Voltage - Short Term
7	Low/ High Voltage - Long Term
8	Phase Issue
9	Additional Feeder
10	Damaged Transformer - Rural
11	Damaged Transformer - Urban
12	Transformer Relocation/Augmentation
13	Additional Transformer
14	Live Fallen Wire
15	Service Line Reloc/Improve - Short Term
16	LT/HT Line Relocation/Improvement- Long Term
17	Meter Sparking/Wire Loose
18	Permanent Rectification of Temporary Fix
19	Tripping (Due to Transformer)
20	Transformer Oil Leakage
21	Leakage of current
22	Damaged Transformer - Independent Consumer
23	Other

C.2 Non-Line Complaints

No.	Complaint Type
1	Meter Position
2	Detection Bill

No.	Complaint Type
3	Delayed Billing (New Connection)
4	Late/Non-Delivery of Bill
5	Electricity Theft
6	Installed Transformer/Meter/Wire Theft
7	Bulk Distribution Theft
8	Electrical Safety
9	Defective Meter (1-phase)
10	Defective Meter (3-phase)
11	Wrong Meter Reading
12	No Meter Reading Taken
13	Excess Billing
14	Under Billing
15	Account Information
16	Bribery/Corruption
17	Delayed Meter Reading
18	Change of Name
19	Other

C.3 Leads / Requests / Others

No.	Complaint Type
1	Additional Transformer
2	Temporary Connection
3	Relocation of Meter
4	New Connection
5	Reconnection
6	Change of Sanctioned Load
7	Electrification
8	Loadshedding Schedule
9	Change of Tariff
10	Replacement Distribution Box
11	Net Metering
12	Apna meter apni reading

No.	Complaint Type
13	Other

C.4 Other

No.	Complaint Type
1	Other

Appendix D — Employee Departmental Complaint Catalog

Employee classification uses Department → Complaint Type, with optional subcategory refinement during implementation. This initial catalog expands the supplied examples across all listed departments and remains subject to MEPCO validation.

D.1 Chief Executive Office (CEO)

No.	Complaint / Request Type
1	Strategic directive clarification
2	Executive correspondence or approval
3	Strategy/KPI tracking issue
4	Inter-directorate coordination
5	Confidential executive support
6	Other

D.2 Company Secretary Office

No.	Complaint / Request Type
1	Board meeting scheduling/material
2	Minutes or resolution record
3	Statutory filing/governance deadline
4	Corporate record access
5	Legal/governance clarification
6	Other

D.3 Internal Audit Section

No.	Complaint / Request Type
1	Audit observation or response

No.	Complaint / Request Type
2	Audit document/evidence request
3	Control weakness or compliance breach
4	Asset verification
5	Audit system/access issue
6	Other

D.4 HR & Administration Directorate

No.	Complaint / Request Type
1	Workplace Harassment
2	Discrimination
3	Payroll & Compensation
4	Benefits & Insurance
5	Leave & Attendance
6	Performance Review
7	Code of Conduct
8	Office Maintenance
9	Communications
10	Office Supplies
11	Physical Security
12	Other

D.5 Finance Directorate

No.	Complaint / Request Type
1	Salary/payment processing
2	Travel or expense reimbursement
3	Budget allocation/approval
4	Vendor payment
5	Ledger/accounting discrepancy
6	Tax or deduction matter
7	Financial report access/error
8	Other

D.6 Commercial/Customer Services Directorate (CSD)

No.	Complaint / Request Type
1	Billing or revenue system issue
2	Consumer service escalation
3	Metering/M&T request
4	New connection/tariff case
5	Recovery or arrears matter
6	Customer record correction
7	Other

D.7 Operations (OP) Directorate

No.	Complaint / Request Type
1	Safety Protocol Violations
2	Field Equipment Shortage
3	Vehicle Maintenance
4	Inventory Management
5	Work Order Discrepancies
6	Feeder/outage coordination
7	Load management or field staffing
8	Other

D.8 Information Technology (IT) Directorate

No.	Complaint / Request Type
1	Hardware
2	Network & Connectivity
3	Enterprise Software (ERP/SAP)
4	Billing Systems
5	Account/Security
6	Digital Portals
7	Other

D.9 Strategic Planning & Engineering Directorate

No.	Complaint / Request Type
1	Network expansion proposal
2	Load forecast/planning data
3	Design/drawing review
4	Estimate or project approval
5	GIS/survey information
6	Project progress coordination
7	Other

D.10 Operations & Maintenance (O&M) T&G Directorate

No.	Complaint / Request Type
1	Grid station equipment fault
2	Transmission line issue
3	Protection/control system issue
4	Scheduled maintenance request
5	SCADA/telecommunication issue
6	Safety or switching coordination
7	Other

D.11 Supply Chain Management Office

No.	Complaint / Request Type
1	Purchase requisition
2	Tender/procurement status
3	Vendor registration/performance
4	Inventory availability
5	Delivery/inspection discrepancy
6	Warehouse record issue
7	Other

D.12 Regional Training Centre (RTC)

No.	Complaint / Request Type
1	Training nomination/enrollment
2	Course schedule
3	Attendance/certification
4	Training material
5	Lab/classroom/facility issue
6	Trainer or evaluation matter
7	Other

D.13 MEPCO Intelligence/Surveillance Unit

No.	Complaint / Request Type
1	Theft/Misuse of Property
2	Corruption/Bribery
3	Safety Incident Reporting
4	Audit & Compliance
5	Electricity theft intelligence
6	Surveillance/investigation request
7	Evidence submission/access
8	Confidential case coordination
9	Other

D.14 Other

No.	Complaint / Request Type
1	Other

Appendix E — Notification Trigger Matrix

Event	Who is notified	Notification purpose
Ticket submitted	Requester and relevant supervisor queue	Confirm submission and show that a new ticket needs attention.
Assigned or reassigned	Requester, new technician and supervisor	Show who is now responsible for the ticket.
Public comment	The other ticket participants	Tell users that a new visible update was added.

Event	Who is notified	Notification purpose
Information requested	Requester	Ask the requester to provide missing details.
Resolved	Requester and supervisor	Ask for review or closure approval.
Closed	Requester and technician	Confirm that the ticket has been completed.
Reopened	Requester, technician and supervisor	Show that the ticket needs more work.
Account suspended	Affected account where policy allows	Explain that access has been restricted.
Announcement	Selected user roles	Share a general system or service notice.

Appendix F — Suggested Project Directory

Path	Purpose
/frontend	React + TypeScript + Vite application, components, pages, API client and UI tests.
/backend	Express + TypeScript API, middleware, modules/services, data access, tests and OpenAPI setup.
/database	Schema/migrations, seed scripts and backup/restore notes if kept outside backend.
/docs	SRS, diagrams, API notes, decisions and presentation source/export.
/docker-compose.yml	Local multi-service environment.
/.env.example	Documented non-secret configuration keys.
/README.md	Setup, run, test, seed, API documentation and troubleshooting guide.
/.github	Pull-request template and optional CI workflow.

Appendix G — Final Validation Checklist

- ☐ MEPCO mentor confirms combined Consumer + Employee scope.
- ☐ Departments, circles, cities and complaint catalogs are validated.
- ☐ Confidential-category visibility and escalation rules are approved.
- ☐ Registration/identity verification method is approved for demonstration and future production.
- ☐ Ticket statuses, priorities, closure/reopening and cancellation rules are approved.
- ☐ Service targets, retention, backup and operational ownership are defined before production use.
- ☐ All twelve mentor-mandated engineering features are demonstrated.
- ☐ Any agreed requirement changes are reflected in the SRS and presentation.

End of Document

Document status

This SRS is ready for discussion, presentation and implementation planning. The remaining MEPCO-specific decisions are clearly listed in Section 16 and Appendix G.